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MOOCS Decision Modeling Course

Decision Analysis

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Decision Analysis: Introduction

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Decision Analysis

- An analytic and systematic approach used to develop an optimal decision strategy when a decision maker is faced with several decision alternatives.

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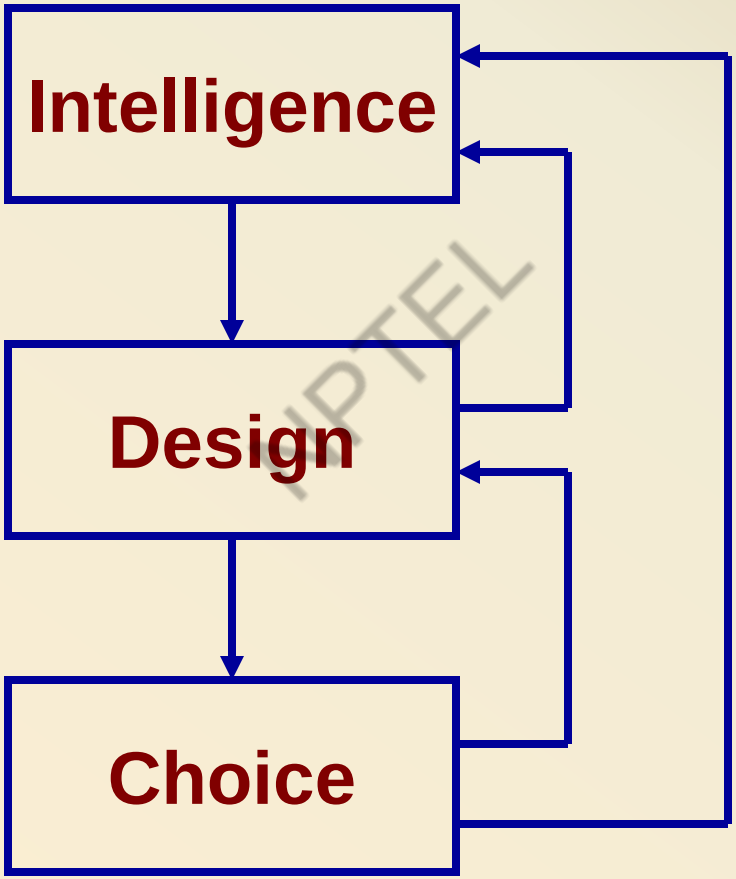
Applications

- Product and service design
- Evaluating financial alternatives
- Marketing decision making
- Capacity planning

Decision Making Process



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Classical Rational Decision Making Model

When faced with a decision situation, Rational or Normative decision makers:

1. Identify problem
2. Identify Decision Criteria
3. Allocate Weights to Criteria
4. Develop Alternatives
5. Analyze Alternatives
6. Select “Best” Choice
7. Implement
8. Evaluate Effectiveness

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Administrative Model Decision Making

- When faced with a decision situation, administrative or descriptive decision makers:
  - Use incomplete and imperfect information
  - Are constrained by bounded rationality
  - Tend to satisfice immediate concerns
  - Finally, choose a decision that may not be optimum but only satisficing.

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Decision Making Environments

Decision Environments of...

Certainty | **Risk** | **Uncertainty**

Low — Moderate — High

Ambiguity and chances of making a bad decision

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Decision Making Under Certainty

Alternatives Known with certainty

Outcomes Known with certainty

Decision-making is therefore straightforward as there is no ambiguity

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Decision Models:

- Linear Programming
- Analytic Hierarchy Process
- EOQ and EPQ Models

Decision Making Under Risk

Alternatives and Outcomes

Known

Alternative → Outcomes

Known only with probability associated with them. However, the probabilities are known.

Decision-making is therefore at risk because of ambiguities due to the probabilities.

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Decision Making Under Uncertainty

Alternatives and Outcomes

Not known with certainty

Alternative → Outcomes

Not known – even the probabilities associated with them are not known.

Decision-making is therefore uncertain because of very high ambiguity.

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Criteria for Decision Making Under Uncertainty

1. Maximax or Optimism Criterion
2. Maximin or Pessimism Criterion
3. Hurwicz or Criterion of Realism
4. Laplace or Equally likely Criterion
5. Savage or Minimax regret Criterion

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Criteria for Decision Making Under Risk

Expected Value Approach: Also known as Expected Monetary Value (EMV) Approach

Steps:

1) Obtain Probability of Occurrence (p_i) for each state of nature.
2) Find the Expected Value for each decision alternative as the sum of (p_i x $payoff_i$) for all the states of nature.
3) Select the decision alternative with the best expected value

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Expected Value of Perfect Information (EVPI)

- EVPI signifies the gain in expected return obtained from knowing with certainty the future state of nature
- EVPI is the maximum amount a decision maker would pay for additional information
- EVPI can be obtained by assuming that a consultant will be able to give perfect information about the true state of nature.
- EVPI will always equal the minimum expected regret

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Estimating Probability by Bayesian Analysis

- Management experience or intuition
- History
- Existing data
- Need to *revise* probabilities based on new data

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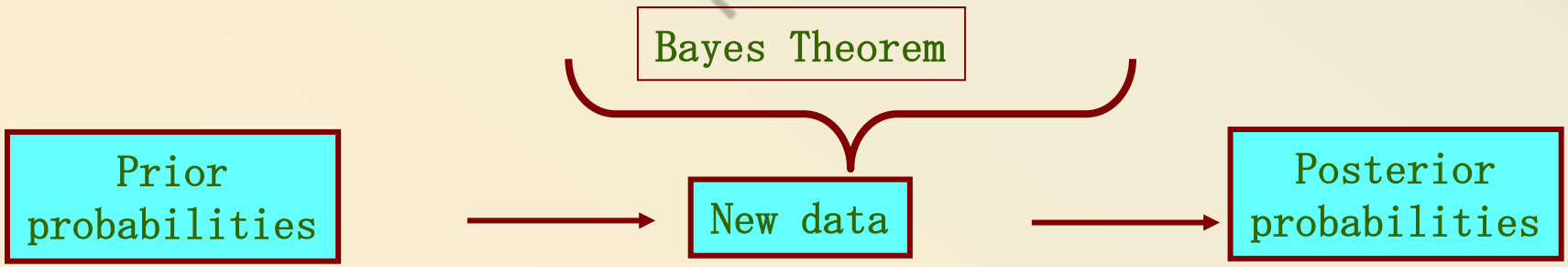
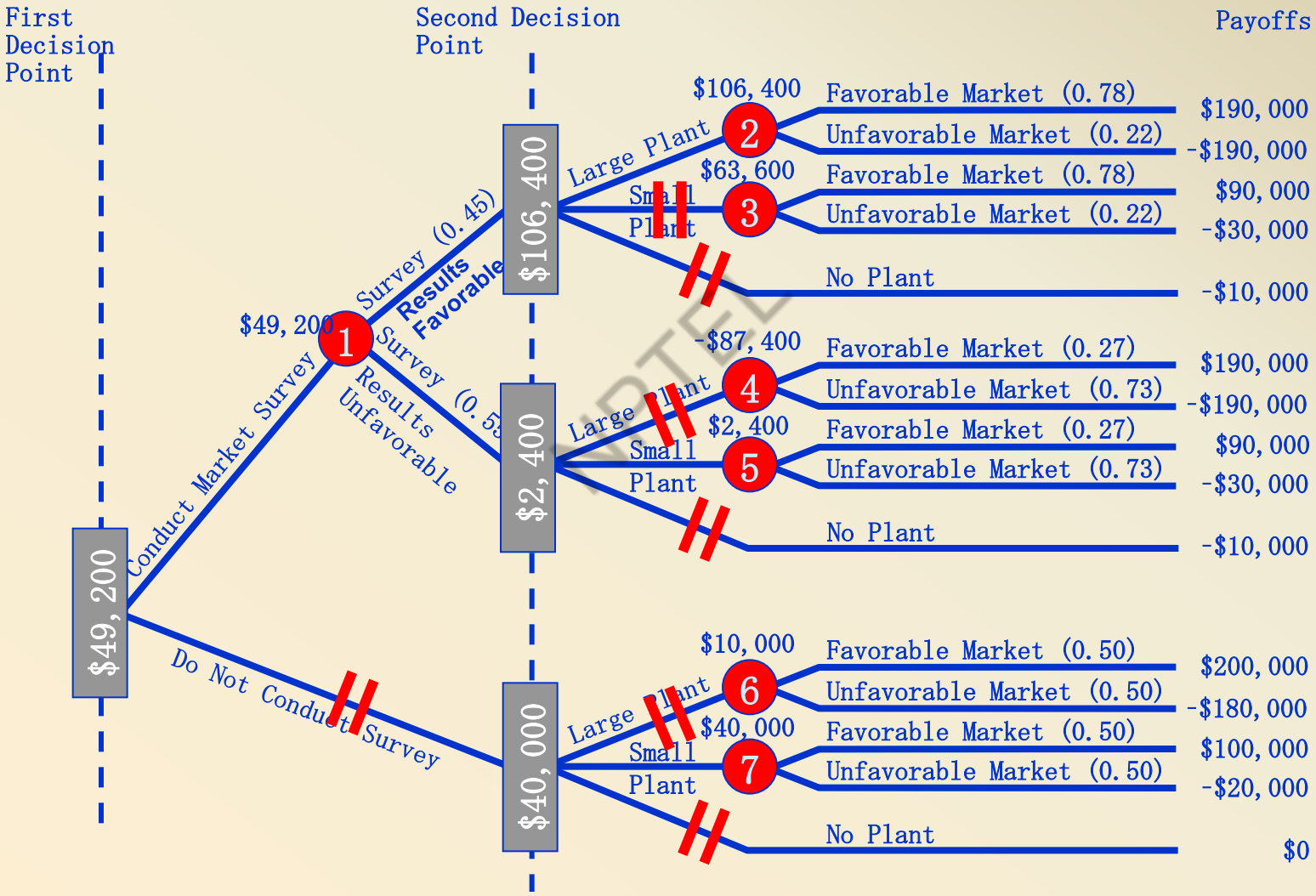


Showing Expected Values on Decision Tree



Payoff Matrix

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Payoff Matrix – An Example

	STATE OF NATURE			
Decision Alternatives	No Market Changes	Favorable market changes	Unfavorable Market changes	Payoff for a Decision Strategy
Redesign on a small scale	30	100	–80	
Rebuild and Refurbish	50	200	–200	
Do nothing	0	30	–50	

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Decision Analysis for Payoff Matrix

- Payoff matrix states of nature and decision strategies may or may not be known with certainty.
- Probabilities of occurrence of outcomes may also be not known.
- Decision analysis therefore may be a combination of decision making under certainty as well as uncertainty

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Criteria for Decision Making Under Uncertainty

1. Maximax or Optimism Criterion
2. Maximin or Pessimism Criterion
3. Hurwicz or Criterion of Realism
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Maximax: Optimism Criterion

- Find the maximum possible payoff for each decision alternative
- Select the decision alternative with the maximum of the above maximum payoffs

- Solve Now:

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	STATE OF NATURE			
Decision Alternatives	No Market Changes	Favorable market changes	Unfavorable Market changes	Payoff for a Decision Strategy
Redesign on a small scale	30	100	–80	
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Do nothing	0	30	–50	

Maximin: Pessimism Criterion

- Find the minimum possible payoff for each decision alternative
- Select the decision alternative with the maximum of the above minimum payoffs

- Solve Now:

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	STATE OF NATURE			
Decision Alternatives	No Market Changes	Favorable market changes	Unfavorable Market changes	Payoff for a Decision Strategy
Redesign on a small scale	30	100	–80	
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Do nothing	0	30	–50	

Hurwicz: Criterion of Realism

- Choose a Degree of Optimism (a). Obtain Maximum Payoff (Max) and Minimum Payoff (Min) for each decision alternative. Obtain Hurwicz Payoffs for each strategy as: a*Max + (1 – a)*Min
- Select the decision alternative with the maximum of the above Hurwicz payoffs
- Solve Now:

	STATE OF NATURE			
Decision Alternatives	No Market Changes	Favorable market changes	Unfavorable Market changes	Payoff for a Decision Strategy
Redesign on a small scale	30	100	–80	
Rebuild and Refurbish	50	200	–200	
Do nothing	0	30	–50	

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Laplace: Equally Likely Criterion

- Find the average value of payoff for each decision alternative
- Select the decision alternative with the maximum of the above average payoff values

- Solve Now:

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	STATE OF NATURE			
Decision Alternatives	No Market Changes	Favorable market changes	Unfavorable Market changes	Payoff for a Decision Strategy
Redesign on a small scale	30	100	–80	
Rebuild and Refurbish	50	200	–200	
Do nothing	0	30	–50	

Savage: Minimax Regret Criterion

- Find the regret matrix (For each payoff under a state of nature, compute difference from maximum payoff). Obtain maximum regret for each decision alternative
- Select the decision alternative with the minimum of the above maximum regrets

- Solve Now:

	STATE OF NATURE			
Decision Alternatives	No Market Changes	Favorable market changes	Unfavorable Market changes	Payoff for a Decision Strategy
Redesign on a small scale	30	100	–80	
Rebuild and Refurbish	50	200	–200	
Do nothing	0	30	–50	

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Decision Making Under Risk

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Criteria for Decision Making Under Risk

Expected Value Approach: Also known as Expected Monetary Value (EMV) Approach

Steps:

1) Obtain Probability of Occurrence (p_i) for each state of nature.
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3) Select the decision alternative with the best expected value

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Example for Expected Value Approach

Decision Alternatives (Prob. of Occurrence)	STATE OF NATURE: No Market Changes (0.3)	Favorable market changes (0.4)	Unfavorable Market changes (0.3)	Expected Value for a Decision Strategy
Redesign on a small scale	30	100	–80	25
Rebuild and Refurbish	50	200	–200	35
Do nothing	0	30	–50	–3

Best Exp. Value

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Example for Expected Value Approach

Decision Alternatives (Prob. of Occurrence)	STATE OF NATURE: No Market Changes (0.3)	Favorable market changes (0.4)	Unfavorable Market changes (0.3)	Expected Value for a Decision Strategy	
Redesign on a small scale	30	100	–80	25	
Rebuild and Refurbish	50	200	–200	35	Best Exp. Value
Do nothing	0	30	–50	–3	

EV (Redesign) : 30x0.3 + 100x0.4 + (-80)x0.3 = 25

EV (Rebuild) : 50x0.3 + 200x0.4 + (-200)x0.3 = 35 Best EV

EV (Do Nothing) : 0x0.3 + 30x0.4 + (-50)x0.3 = –3

Best Decision Strategy: Rebuild and Refurbish

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Sensitivity Analysis

Consider a smaller payoff matrix problem. - In this problem, the decision strategy "Constructing a small plant" shows the largest EV.

We need to carry out sensitivity analysis for this decision problem:

	STATE OF NATURE		
ALTERNATIVE	FAVORABLE MARKET ($)	UNFAVORABLE MARKET ($)	EV ($)
Construct a large plant	200,000	–180,000	10,000
Construct a small plant	100,000	–20,000	40,000
Do nothing	0	0	0
Probability	0.50	0.50	

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Sensitivity Analysis

For the example Problem

Let P = probability of a favorable market

$(1 - P)$ = probability of an unfavorable market

EV(Large Plant) $= 200{,}000P - 180{,}000)(1 - P)$
$= 380{,}000P - 180{,}000$

EV(Small Plant) $= 100{,}000P - 20{,}000)(1 - P)$
$= 120{,}000P - 20{,}000$

EV(Do Nothing) $= 0P + 0(1 - P) = 0$
$= 0$

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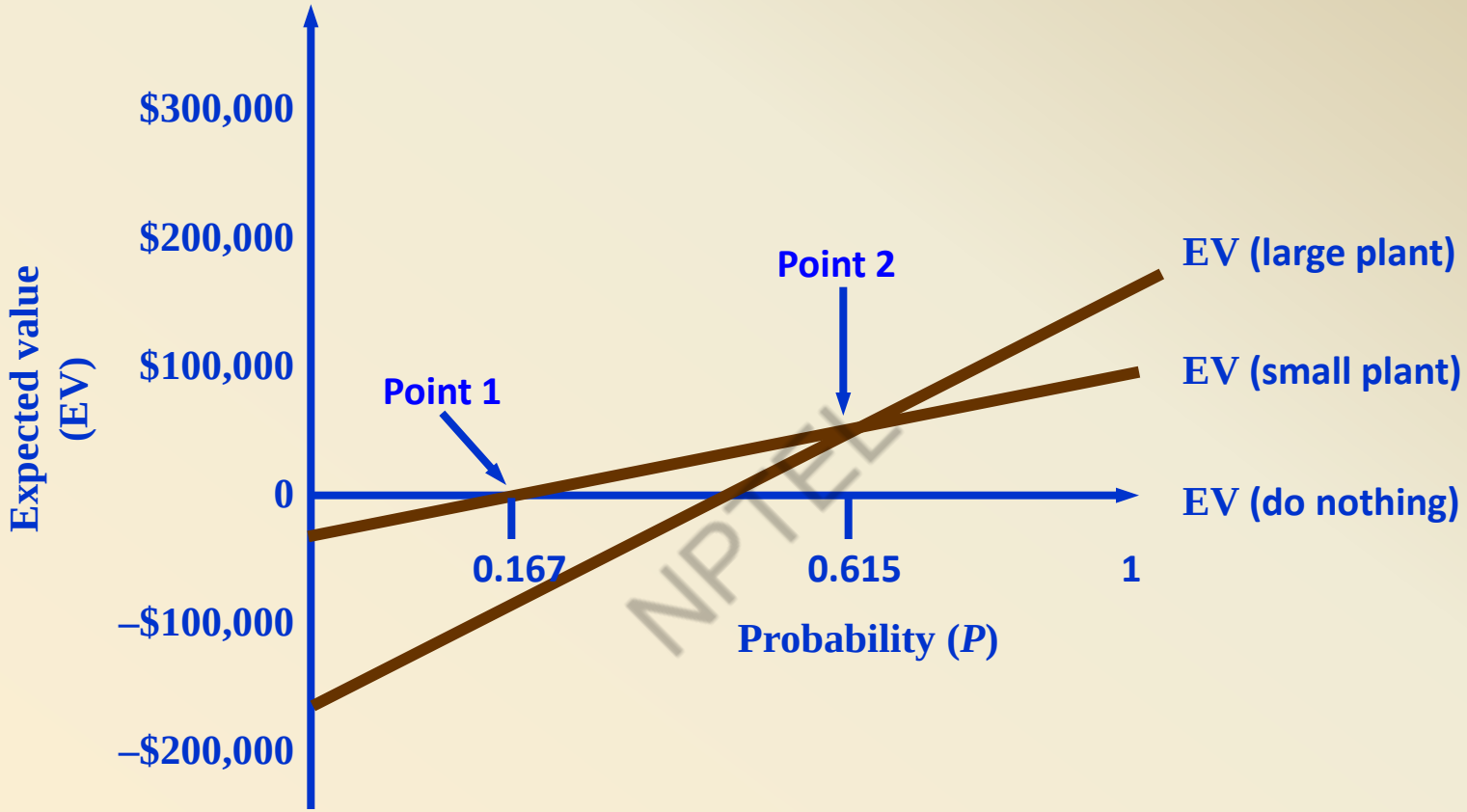
Sensitivity Analysis



EV(Large Plant) = 380,000*P* – 180,000
EV(Small Plant) = 120,000*P* – 20,000
EV(Do Nothing) = 0

Sensitivity Analysis

Point 1:

EV(do nothing) = EV(small plant)

$$0 = \$120{,}000P - \$20{,}000 \qquad P = \frac{20{,}000}{120{,}000} = 0.167$$

Point 2:

EV(small plant) = EV(large plant)

$$\$120{,}000P - \$20{,}000 = \$380{,}000P - \$180{,}000$$

$$P = \frac{160{,}000}{260{,}000} = 0.615$$

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BEST ALTERNATIVE	RANGE OF P VALUES
Do nothing	Less than 0.167
Construct a small plant	0.167 – 0.615
Construct a large plant	Greater than 0.615

Value of Information

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Expected Value of Perfect Information (EVPI)

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Expected Value of Perfect Information (EVPI)

EVPI = ERPI – best EREV

Where,

EREV: Expected Return of the EV approach

ERPI: Expected Return with Perfect Information

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ERPI= (best payoff for first state of nature) x (probability of first state of nature) + (best payoff for second state of nature) x (probability of second state of nature) + … + (best payoff for last state of nature)x (probability of last state of nature

Expected EVPI

Alternative	State of Nature		Expected Value (EV)
	Favorable Market ($)	*Unfavorable Market ($)*	
Construct a large plant	200,000	-180,000	10,000
Construct a small plant	100,000	-20,000	40,000
Do nothing	0	0	0
Probability	**0.5**	**0.5**	
Perfect Information	200,000	0	ERPI = 100,000

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Largest EV

Compute ERPI

State of Nature: Favorable Market, Best Payoff: 200,000

State of Nature: Unfavorable Maket, Best Payoff: 0

ERPI = (200,000)(0.5) + (0)(0.5) = $100,000

Compute EVPI

EVPI = ERPI – max EREV = $100,000 – $40,000 = $60,000 - the most we should pay for additional information

Probability Concepts

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Probability Concepts

Probability: Likelihood or the chance that an uncertain event would happen.

Three approaches to determine probability of an event:

Classical Theory

P(E) = No. of favorable outcomes/Total Possible Outcomes

Relative Frequency **(Requires data collection)**

P(E) = No. who purchased product/Total Number

Personalistic Approach

Degree of belief

In all of these, we have: 0 <= P (E) <= 1

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Probability Concepts

Mutually Exclusive Events:

P(E1 or E2) = P(E1 U E2) = P(E1) + P(E2)

Overlapping Events:

P(E1 U E2) = P(E1) + P(E2) – P (E1 ∩ E2)

Example

	Cat1	Cat2	Cat3	Total
Male	250	50	350	650
Female	350	200	650	1200

Find the chance that a randomly selected employee:

a) female, b) Category 2, c) male and Category 3

d) Employee Cat1 or Cat3, e) Female or Rank3

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Conditional Probability

Let us start with tossing an unbiased die once and obtain number returned.

Let E: event when a 6 is returned.

Let F: event when an even number is returned.

Now suppose we would like to know,

the probability of event E given that event F has occurred.

In other words, if it is known that an even face has turned up, what is the probability that it is a 6?

New (reduced) sample space is S = {2, 4, 6} Outcomes of F

Conditional Probability of *E* given *F* is the probability that an event, *E*, will occur given that another event, *F*, has occurred:

P(E occurs given that *F* occurs) = 1/3

i.e. P(E|*F*) = 1/3

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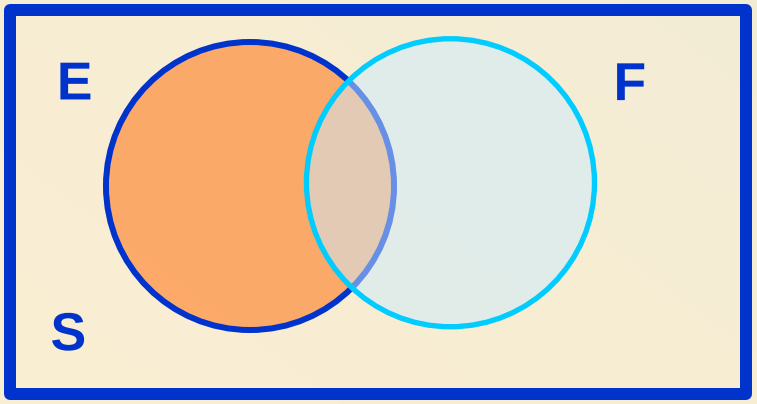
Conditional Probability

Conditional Probability of *E* given *F* is the probability that an event, *E*, will occur given that another event, *F*, has occurred:

P(E occurs given that *F* occurs) = P(E|*F*)

In other words, Conditional Probability of E given F is the ratio of Joint Probability of E and F divided by Probability of F.

$$P(E \mid F) = \frac{P(E \cap F)}{P(F)} \quad \text{if} \quad P(F) \neq 0$$



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Conditional Probability

- **If the outcomes of an experiment are equally likely, then**

$$P(E \mid F) = \frac{\text{number of outcomes in } E \cap F}{\text{number of outcomes in } F}$$

Example

	Cat1	Cat2	Cat3	Total
Male	250	50	350	650
Female	350	200	650	1200
Total	600	250	1000	1850

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Find the chance that a randomly selected employee:

a) Female given category 1, b) male given Category 2

a) P(Female|Cat1) = 350/600

b) P(Male|Cat2) = 50/250

Conditional Probability

- **Conditional Probability can be rewritten as:**

$$P(E \mid F) = \frac{P(E \cap F)}{P(F)} \quad \text{to} \quad P(E \cap F) = P(E \mid F) * P(F)$$

Example

	Cat1	Cat2	Cat3	Total
Male	250	50	350	650
Female	350	200	650	1200
Total	600	250	1000	1850

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Find the chance that a randomly selected employee:

a) Female given category 1, b) male given Category 2

a) P(Female|Cat1) = 350/600

b) P(Male|Cat2) = 50/250

Independent Events

If the probability of the occurrence of event E is the same whether or not event F occurs, then events E and F are independent of one another.

$$P(E \mid F) = P(E)$$

i.e. E and F are independent events.

Since,

$$P(E \cap F) = P(E \mid F) * P(F)$$

If and only if E and F are independent events, we have:

$$P(E \cap F) = P(E)P(F)$$

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Independent vs Mutually Exclusive Events

- **Independent Events vs. Mutually Exclusive Events (Disjoint Events)**

- **If two events are Independent,**

$$P(E \mid F) = P(E)$$

$$P(E \cap F) = P(E)P(F)$$

- **If two events are Mutually Exclusive, then they do not share common outcomes**

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THANK YOU

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